PLC-BASED MULTI-SENSOR CONDITION MONITORING OF INDUSTRIAL MOTORS USING ELECTRICAL CURRENT AND VIBRATION SIGNALS

Takunda Pambayi

Harare Institute of Technology, Harare, Zimbabwe

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Research status: design and validation framework with worked engineering calculations. No unverified experimental results are presented as measured data.

Abstract

Industrial induction motors are essential prime movers in manufacturing plants, and failure of a single motor can interrupt a complete process section. Conventional preventive maintenance is commonly based on calendar intervals, operating hours and technician experience. Although useful, this approach does not continuously reflect the actual condition of the machine. This paper develops a practical PLC-based multi-sensor condition-monitoring framework that combines motor electrical current and mechanical vibration measurements. The proposed architecture separates sensing, signal conditioning, feature extraction, PLC decision-making and HMI/SCADA visualization. Current RMS, phase-current imbalance, vibration RMS, peak value, crest factor and selected frequency-domain components are used as condition indicators. A transparent sensor-fusion method is introduced in which normalized electrical and vibration indicators are combined before a normal, warning or critical state is assigned. The paper also provides worked calculations for RMS current, current imbalance, vibration RMS, crest factor, FFT frequency resolution and a weighted condition score. These calculations demonstrate how raw measurements can be converted into engineering quantities suitable for PLC implementation. An experimental validation procedure is defined for healthy operation, controlled mechanical faults, increased load and selected electrical abnormalities, with detection rate, false-alarm rate, detection latency and monitoring availability proposed as evaluation metrics. The approach is intended for industrial applications where PLC and HMI infrastructure already exists and where a specialist condition-monitoring platform may be economically or technically impractical. The contribution is therefore not a claim of completed fault-classification performance, but a reproducible engineering framework that can be implemented and experimentally validated on industrial motor-driven equipment.

Keywords: PLC; condition monitoring; induction motor; motor current signature analysis; vibration analysis; multi-sensor fusion; predictive maintenance; industrial automation.

1. Introduction

Electric motors are fundamental to modern production systems because they provide the mechanical power required by pumps, compressors, fans, conveyors and process machinery. The reliability of these machines is consequently linked to production availability, maintenance cost and process continuity. Motor condition monitoring has developed from periodic inspection toward online measurement of electrical, mechanical and thermal variables. Current signature analysis is particularly attractive because stator current can be measured without direct access to the rotating shaft, whereas vibration provides a direct indication of mechanical behaviour (Benbouzid, 2000; Gangsar and Tiwari, 2020).

The limitation of relying on a single measurement is that different fault mechanisms can produce overlapping symptoms. A rise in vibration may be associated with imbalance, misalignment, looseness or bearing deterioration, while an increase or distortion in current may result from loading, supply imbalance, electrical faults or electromechanical coupling. Research has therefore investigated combinations of current and vibration information for online diagnosis (Yang and Kim, 2006). More recent reviews also show continued development of current-signature methods, including approaches for variable-speed operation where fixed spectral features can become difficult to interpret (Niu, 2023; Yakhni et al., 2023).

Industrial automation provides an opportunity to move part of this diagnostic process into equipment that already exists in many factories. PLCs are routinely used for motor starting, sequencing, interlocking, protection signals and process supervision. Integrating condition indicators with the PLC therefore reduces the separation between production control and maintenance information. The proposed work addresses this practical implementation problem by developing a PLC-centred architecture in which current and vibration information are processed into transparent condition indicators.

1.1 Problem Statement

Many industrial maintenance programmes still depend on fixed inspection intervals or respond after abnormal equipment behaviour has become visible. Such approaches can miss developing faults between inspections and

may result either in premature maintenance or in unexpected breakdowns. Specialist condition-monitoring systems can provide more information, but their cost, integration requirements and dependence on specialist software may limit adoption in smaller manufacturing facilities. The engineering problem addressed in this paper is therefore the design of a practical, low-complexity monitoring system that combines electrical and mechanical evidence and presents an interpretable decision through a PLC and HMI.

1.2 Aim and Objectives

The aim is to develop and define a PLC-based multi-sensor condition-monitoring system for industrial induction motors using electrical current and vibration signals.

- Develop a sensing architecture for simultaneous acquisition of motor current and vibration.
- Define time-domain and frequency-domain features suitable for condition assessment.
- Develop a transparent PLC decision and sensor-fusion method for normal, warning and critical states.
- Develop an HMI/SCADA presentation structure for trends, alarms and maintenance information.
- Define a controlled experimental procedure for comparing current-only, vibration-only and fused monitoring.
- Establish objective performance measures including detection rate, false-alarm rate and detection latency.

2. Literature Review and Research Gap

Condition monitoring methods for induction motors include temperature measurement, insulation testing, current signature analysis, vibration analysis, magnetic-flux monitoring, acoustic techniques and combinations of these signals. Gangsar and Tiwari (2020) reviewed signal-based monitoring and emphasized the value of both vibration and current measurements for electrical and mechanical fault diagnosis. Benbouzid (2000) established the basis of motor current signature analysis as a means of identifying abnormal conditions from the spectral content of stator current. These studies support the use of current as an online, non-invasive measurement channel.

Vibration monitoring provides complementary mechanical information. Bearing defects, imbalance, misalignment and looseness can change both the magnitude and frequency content of the vibration signal. Yang and Kim (2006) demonstrated a data-fusion approach using vibration and current features for induction-motor fault diagnosis, providing a direct precedent for combining the two measurement domains. The advantage of the present work is to translate this multi-sensor principle into an automation-oriented PLC architecture rather than treating the diagnostic system only as an offline signal-processing exercise.

Variable-speed operation introduces an additional challenge. Yakhni et al. (2023) note that transient motor-current signatures become more difficult to interpret when the operating point changes. VFD switching, load variation and speed changes can alter the measured spectrum. Therefore, the proposed system does not depend on one universal frequency peak. Instead, the operating state, baseline condition and multiple features are considered together.

2.1 Comparison of Monitoring Approaches

Approach	Main measurement	Main strength	Main limitation
Manual inspection	Observation / physical inspection	Low cost and familiar	Intermittent and operator-dependent
Temperature	Motor temperature	Useful for thermal overload	May not identify early mechanical faults
Vibration	Accelerometer	Strong sensitivity to mechanical faults	Mounting and interpretation are important
MCSA	Stator current	Non-invasive and suitable for online monitoring	Affected by load, speed and power electronics

Multi-sensor	Current + vibration	Combines complementary evidence	More sensing and processing required
Proposed PLC system	Current + vibration + PLC state	Direct integration with automation	Requires validated thresholds and conditioning

2.2 Research Gap

The literature establishes that current and vibration contain complementary information, but there remains a practical implementation gap between advanced diagnostic research and the control architecture available on industrial floors. Many systems rely on specialist acquisition hardware, offline analysis or computationally intensive algorithms. The proposed research focuses on the intermediate engineering layer: extracting a small but meaningful set of indicators and implementing transparent decision logic at PLC level. This creates a pathway from conventional preventive maintenance toward condition-based maintenance without requiring immediate deployment of a complex machine-learning platform.

3. Proposed System Architecture

The system is organised into five functional layers: sensing, signal conditioning, feature extraction, PLC decision-making and operator visualization. The architecture is designed so that the PLC receives reliable condition indicators rather than attempting to process every high-frequency raw vibration sample. This is important because a conventional PLC is primarily a real-time control device, whereas high-resolution vibration processing can require significantly higher sampling and computational resources.

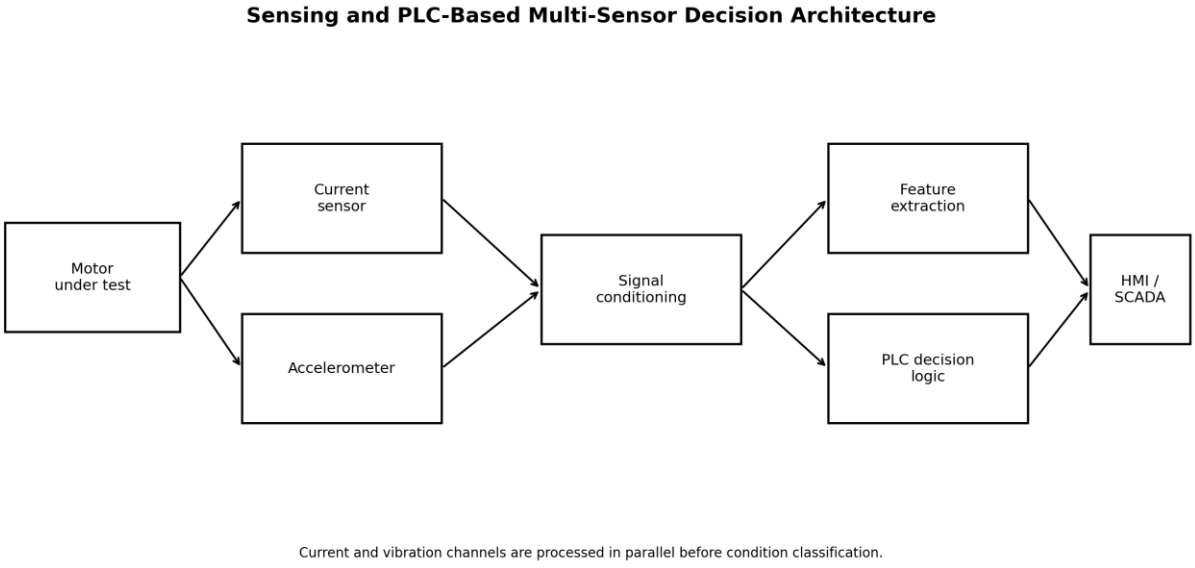


Figure 1. Proposed PLC-based multi-sensor condition-monitoring architecture. The figure represents the proposed design and is not a measured experimental result.

Current is measured using a suitable current transformer or Hall-effect sensor, depending on the motor rating and electrical arrangement. Vibration is measured using an accelerometer mounted consistently on the motor housing, preferably near the bearing location and in a defined radial direction. Sensor signals are isolated, filtered and scaled before analogue-to-digital conversion. Where the required vibration bandwidth is beyond the PLC analogue-input capability, an edge acquisition module can calculate features such as RMS, peak and spectral indicators before sending them to the PLC.

3.1 Sensor Placement and Measurement Discipline

Sensor mounting is an important part of the experimental method because vibration amplitude depends on location, mounting stiffness, direction and the surrounding mechanical structure. Measurements should therefore

be taken from the same location and direction throughout the experiment. ISO 20816-1 provides general guidance on measurement and evaluation of machine vibration and emphasizes the importance of measurement conditions, quantities and operating state (ISO, 2016).

Variable	Recommended measurement discipline	Purpose
Phase current	Same phase reference or all three phases where available	Electrical loading and imbalance
Vibration	Fixed radial bearing-housing position	Mechanical condition
Speed/VFD frequency	Logged with every feature window	Operating-point reference
Load	Recorded or controlled during testing	Separates fault effects from load effects
Sampling rate	Selected from highest frequency of interest	Avoids aliasing
Window length	Fixed for each test condition	Ensures comparable features

4. Signal Processing and Engineering Calculations

The monitoring system converts sampled signals into numerical features that can be interpreted by the PLC. The following calculations are central to the proposed method. Worked examples are included using explicitly illustrative data; they are not claimed to be measurements from the author's motor test rig.

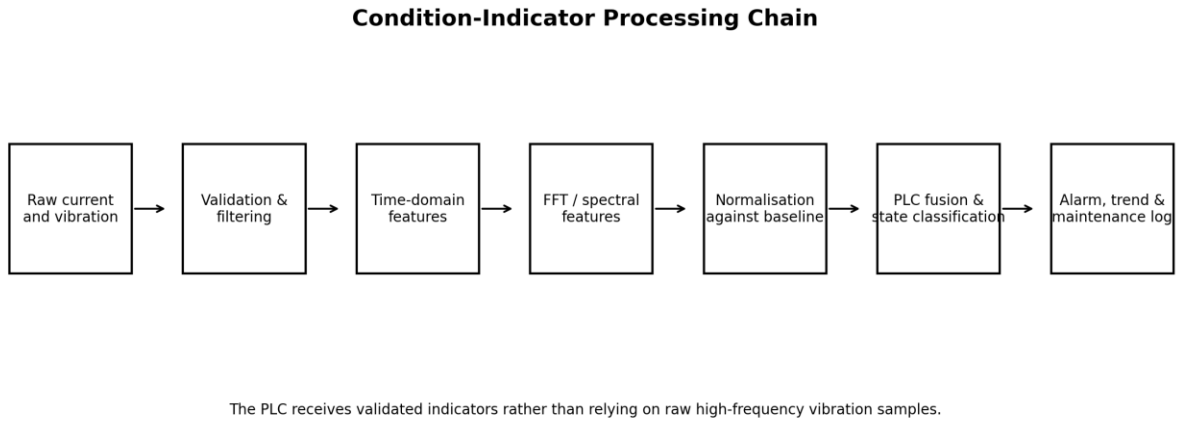


Figure 2. Proposed signal-processing chain from raw measurements to maintenance-oriented PLC decisions.

4.1 Current RMS

For a sampled current waveform containing N samples, the RMS current is calculated as:

$$I_RMS = \sqrt{[(1/N)\ \Sigma(i_k^2)]},\quad k = 1, 2, \ldots, N$$

Consider the illustrative five-sample sequence 6.2, 6.5, 6.1, 6.4 and 6.3 A. The mean current is 6.30 A and the calculated RMS value is approximately 6.302 A. The RMS value provides a compact representation of current loading, but it should not be interpreted as a fault indicator by itself because a healthy motor can legitimately draw different current at different loads.

4.2 Three-Phase Current Imbalance

For three-phase measurements, a simple percentage current-imbalance indicator can be defined as the maximum deviation from the average divided by the average:

$$I_imbalance\ (\%) = [\max|I_phase - I_avg|\ /\ I_avg] \times 100$$

As an illustrative example, if the three measured phase currents are 6.2, 6.4 and 6.0 A, the average is 6.20 A and the maximum deviation is 0.20 A. The resulting imbalance is approximately 3.23%. This calculation can be implemented in the PLC using three analogue channels and a small arithmetic routine.

4.3 Vibration RMS and Crest Factor

Vibration RMS represents the overall energy of the measured acceleration signal. For N samples:

$$V_RMS = \sqrt{[(1/N)\ \Sigma(x_k^2)]}$$

The crest factor provides an additional indicator of impulsive behaviour:

$$\text{Crest Factor} = \text{Peak}\ |x_k|\ /\ V_RMS$$

Using the illustrative vibration samples [0.16, −0.19, 0.21, −0.18, 0.17, −0.20, 0.15, −0.22] g gives V_RMS ≈ 0.186 g and a crest factor of approximately 1.18. An increase in crest factor over a motor-specific baseline can be useful when developing bearing-fault indicators because impulsive events may increase peak acceleration more rapidly than overall RMS.

4.4 Kurtosis

Kurtosis can be used to identify departures from an approximately Gaussian vibration distribution. A simplified population form is:

$$K = [\ (1/N)\ \Sigma(x_k - \mu)\text{^}4\ ]\ /\ \sigma\text{^}4$$

Because kurtosis is sensitive to outliers, it should be interpreted with adequate sample length and consistent sensor mounting. It is proposed here as a supplementary feature rather than a stand-alone alarm variable.

4.5 Fast Fourier Transform and Frequency Resolution

The FFT converts a sampled signal from the time domain into frequency-domain components. For a sampling frequency F_s and transform length N, the nominal frequency-bin spacing is:

$$\Delta f = F_s\ /\ N$$

For an illustrative acquisition system operating at 10 kHz with N = 2048 samples, Δf = 4.883 Hz. This calculation is important when selecting the acquisition parameters because insufficient frequency resolution can make neighbouring spectral components difficult to separate. The sampling frequency must also be sufficiently high relative to the highest frequency of interest to avoid aliasing.

4.6 Normalised Condition Indicators

To combine current and vibration measurements, each feature can be normalised against a healthy baseline. A simple engineering form is:

$$C = \text{clip}[(X_c - X_base)\ /\ (X_warn - X_base),\ 0,\ 1]$$

$$V = \text{clip}[(X_v - V_base)\ /\ (V_warn - V_base),\ 0,\ 1]$$

Here C and V are normalized electrical and vibration indicators. The clip function restricts the value to the interval 0–1 for the basic implementation. Motor-specific baselines are preferred because motor size, mounting, speed and load affect the measured values.

4.7 Sensor-Fusion Score

A transparent weighted fusion score can then be calculated as:

$$S = w_c\ C + w_v\ V,\quad \text{where } w_c + w_v = 1$$

As a worked example only, if C = 0.65, V = 0.85, w_c = 0.40 and w_v = 0.60, then S = 0.77. If an engineering validation programme defined S < 0.60 as normal, 0.60–0.80 as warning and S > 0.80 as critical, the worked

example would fall in the warning band. These limits are illustrative and must not be transferred to a real motor without validation.

5. PLC Decision Logic and HMI Design

The PLC is responsible for deterministic classification, persistence timing, alarm handling and communication with the HMI. The design deliberately separates diagnostic warning thresholds from protective trip functions. A condition-monitoring alarm should not automatically become a motor trip unless the threshold has been validated for that specific machine and integrated with the approved protection philosophy.

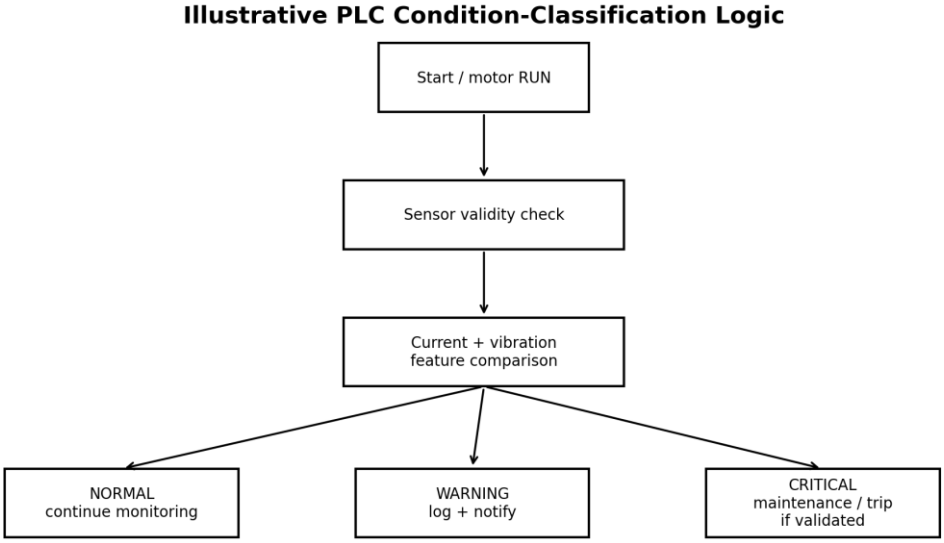


Figure 5. Illustrative PLC condition-classification logic. The final trip action must be validated separately from diagnostic alarms.

Condition state	Typical trigger	PLC/HMI response
Normal	All valid indicators below warning limits	Continue monitoring; green status
Warning	One or more indicators above warning level for persistence period	Alarm, trend logging and maintenance notification
Critical	Multiple abnormal indicators or validated severe condition	Urgent maintenance alarm; trip/interlock only where justified
Sensor fault	Missing, saturated or implausible signal	Sensor fault alarm; inhibit unreliable diagnosis

5.1 Persistence and False-Alarm Control

Industrial measurements can contain short disturbances caused by switching, load changes or mechanical shocks. A practical PLC algorithm should therefore use persistence rather than reacting to one sample. For example, a warning may require the abnormal condition to remain present for several consecutive feature windows. The persistence time should be selected from the machine dynamics and the required response time rather than chosen arbitrarily.

5.2 HMI Information Structure

- Motor identification and running status.
- Current RMS and, where available, three-phase current imbalance.

- Vibration RMS, peak and crest factor.
- Condition state and alarm status.
- Trend history for the main condition indicators.
- Last alarm, acknowledgement and maintenance action.
- Sensor-health indication to prevent decisions based on invalid data.

6. Experimental Validation Method

The current manuscript is intentionally structured as a reproducible experimental design rather than presenting unmeasured values as results. A laboratory motor test rig or an approved industrial motor can be used. The motor should be operated first in a healthy state over a defined operating envelope, after which controlled abnormal conditions can be introduced one at a time. Electrical fault creation should only be performed using a safe laboratory arrangement under competent supervision.

6.1 Experimental Stages

- Record healthy current and vibration data at no-load and representative load points.
- Record motor speed or VFD frequency simultaneously with the sensor data.
- Calculate baseline mean, standard deviation and selected percentiles for each feature.
- Introduce one controlled abnormal condition at a time, such as imbalance, misalignment or increased load.
- Repeat measurements at more than one operating point to test robustness.
- Compare current-only, vibration-only and fused decisions.
- Measure detection rate, false-alarm rate, detection latency and monitoring availability.
- Transfer validated thresholds and logic to the PLC and repeat the test in real time.

6.2 Suggested Test Matrix

Test case	Operating condition	Primary evidence	Secondary evidence	Expected use
T1	Healthy / low load	Stable current	Stable vibration	Baseline
T2	Healthy / rated load	Current increase consistent with load	Vibration within baseline	Load compensation
T3	Controlled imbalance	Vibration at rotational component	Current modulation	Mechanical detection
T4	Controlled misalignment	Radial/axial vibration change where measured	Current deviation	Mechanical-electrical coupling
T5	Increased mechanical load	Current RMS increase	Load-related vibration	Distinguish load from fault
T6	Electrical abnormality under safe test setup	Current imbalance/distortion	Possible vibration response	Electrical diagnosis
T7	Sensor disconnection/saturation	Invalid signal	Independent channel status	Sensor-fault handling

7. Illustrative Analysis and Performance Metrics

The figures in this section are demonstrations of the analysis workflow. They are generated from illustrative mathematical signals and are not measurements from the author's test rig. Replacing them with logged experimental data is a required step before journal submission.

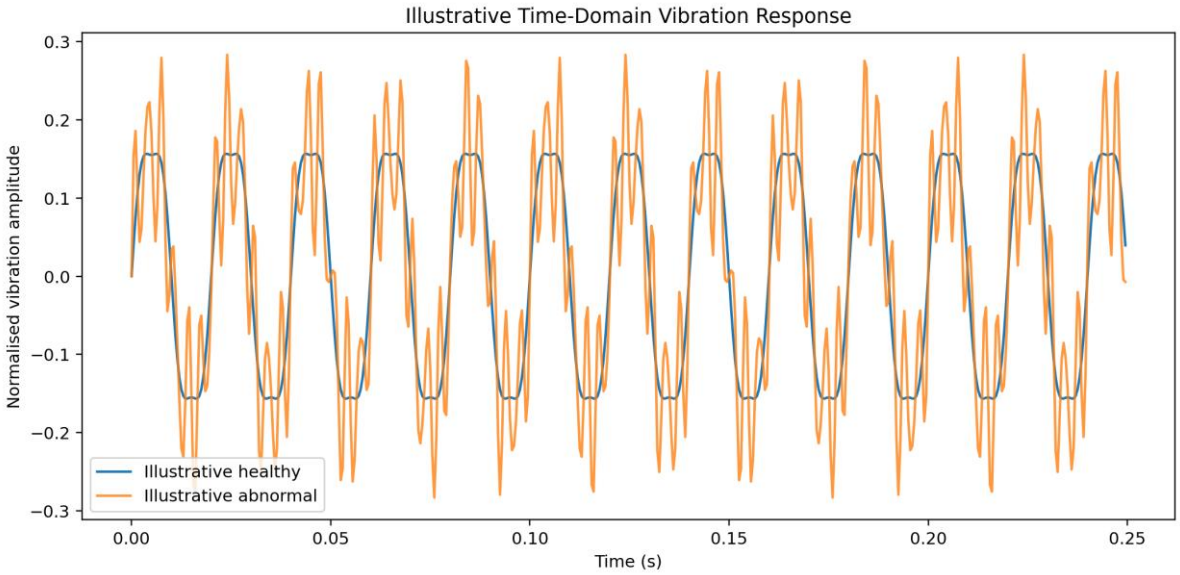


Figure 3. Illustrative time-domain vibration signals used only to demonstrate the distinction between a stable waveform and an abnormal waveform.

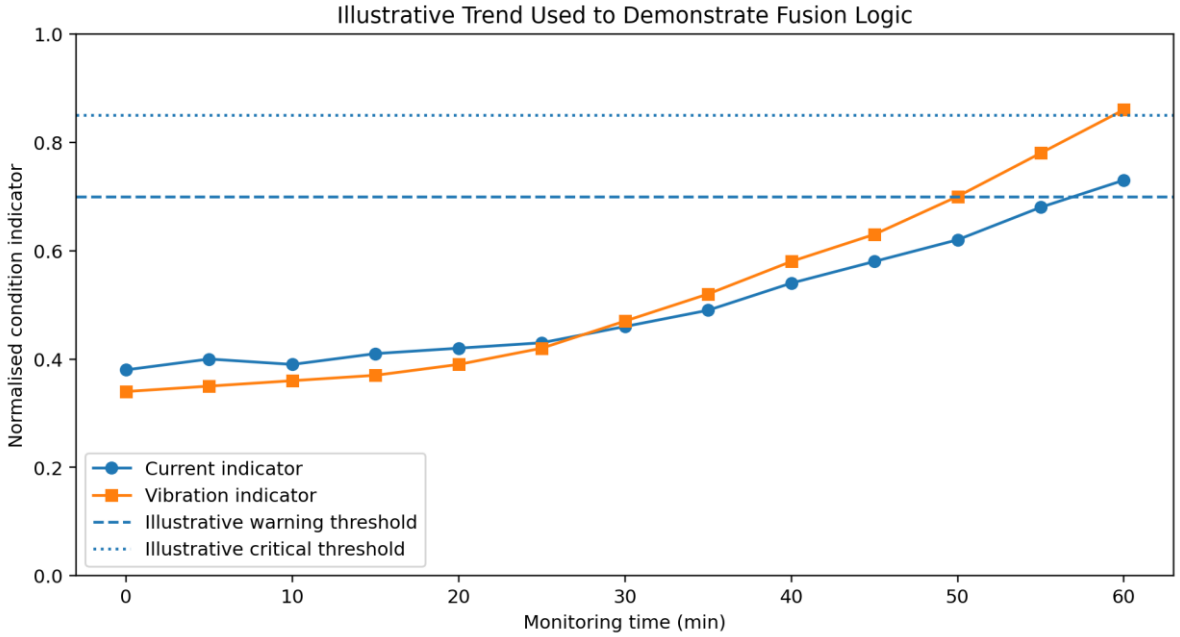


Figure 4. Illustrative normalized current and vibration indicators demonstrating the proposed warning/critical decision concept. Values are not experimental measurements.

7.1 Detection Metrics

Once the experiment is completed, each monitoring window can be classified as normal or abnormal and compared with the known test condition. Detection rate, precision, false-alarm rate and latency should be calculated from the resulting confusion matrix.

$$\text{Detection rate} = TP / (TP + FN)$$

$$\text{False-alarm rate} = FP / (FP + TN)$$

$$\text{Precision} = TP / (TP + FP)$$

$$\text{Detection latency} = t_alarm - t_fault_onset$$

These measures should be calculated separately for each fault class and operating condition rather than reporting one combined number only. A system that performs well at constant speed but generates excessive alarms during VFD operation should not be treated as equally reliable across the entire operating envelope.

7.2 Worked Example of Classification Metrics

To demonstrate the calculation method only, assume a hypothetical validation set contains 20 abnormal windows and 30 healthy windows. If 18 abnormal windows are correctly detected, 2 abnormal windows are missed, and 3 healthy windows generate false alarms, then TP = 18, FN = 2, FP = 3 and TN = 27. The illustrative detection rate is therefore 90.0%, while the false-alarm rate is 10.0% and precision is 85.7%. These numbers are mathematical examples and must not be reported as experimental performance.

7.3 Monitoring Availability

$$\text{Monitoring availability (\%)} = [\text{valid monitoring time} / \text{scheduled monitoring time}] \times 100$$

This metric is important for an industrial deployment because a theoretically accurate algorithm is of limited practical value if sensor faults, communication interruptions or data-quality problems frequently remove it from service.

8. Discussion

The proposed architecture combines electrical and mechanical evidence rather than assuming that one measurement can explain every motor fault. Current information is valuable because it can be measured non-invasively and is related to electromagnetic loading, while vibration responds more directly to mechanical phenomena. The complementary nature of the signals is consistent with published research on combined current and vibration diagnosis (Yang and Kim, 2006; Gangsar and Tiwari, 2020).

The PLC-centred implementation is particularly relevant to manufacturing environments because PLCs are already connected to motors, drives, protection devices and operator interfaces. The proposed architecture can therefore be added as a condition-monitoring layer without replacing the existing process-control system. Where raw vibration processing exceeds PLC capability, the architecture allows feature extraction to be performed at the edge while leaving deterministic alarm logic in the PLC.

An important engineering issue is threshold selection. A single universal vibration or current limit can produce poor results because machines operate under different loads, speeds, mounting conditions and process duties. ISO 20816-1 emphasizes that vibration evaluation depends on defined measurement conditions and operating state (ISO, 2016). Similarly, current signatures can change with variable speed and transient operation, which supports the use of speed-aware or operating-state-aware baselines (Yakhni et al., 2023).

The approach also offers a staged pathway toward predictive analytics. A first implementation can use transparent rules and trends. Once sufficient labelled data are accumulated, statistical models or machine-learning methods can be evaluated against the rule-based baseline. This avoids introducing artificial intelligence before the measurement system, data quality and maintenance workflow are reliable.

8.1 Practical Industrial Benefits

- Earlier visibility of abnormal motor behaviour.
- Reduced dependence on intermittent manual inspection for continuously operating assets.
- Better planning of maintenance interventions using trend information.
- Use of existing PLC/HMI infrastructure rather than a completely separate monitoring platform.
- Structured historical data for future predictive-maintenance development.
- Potential reduction in unplanned downtime when alarms are acted upon appropriately.

8.2 Limitations

The principal limitation of the present manuscript is that the experimental campaign has not yet been completed; therefore, no numerical detection accuracy is claimed. Other limitations include sensor mounting sensitivity, environmental vibration, load dependence, VFD switching effects and the computational limits of conventional PLCs. Thresholds must be validated separately for each motor or motor class. Condition monitoring also provides evidence of abnormal behaviour rather than automatically proving the physical root cause. Maintenance personnel must therefore confirm the fault before major repair or replacement.

9. Implementation Considerations for Industrial Deployment

For a manufacturing plant, the first deployment should focus on critical motor-driven assets where downtime has a clear operational consequence. A practical pilot can begin with one motor, one current-sensing channel and one accelerometer, followed by expansion to multiple assets after the baseline and alarm logic have been validated.

Implementation item	Recommended engineering decision
Motor selection	Select a critical motor with known operating history and stable access
Current sensor	Use a properly rated isolated CT or Hall-effect sensor
Vibration sensor	Use an industrial accelerometer with repeatable mounting
Sampling	Set according to the highest diagnostic frequency of interest
Feature layer	Compute RMS, peak, crest factor and selected spectral indicators
PLC	Use analogue acquisition plus deterministic arithmetic/logic
HMI	Display condition state, trends and alarm history
Data storage	Store timestamp, motor ID, operating state and feature values
Validation	Calibrate thresholds against healthy baseline and controlled faults

IEC 60034-1 specifies general requirements relating to rating and performance of rotating electrical machines; the latest IEC publication is IEC 60034-1:2026. Motor-specific protection, installation and test requirements should therefore be checked against the applicable equipment documentation and standards before field implementation (IEC, 2026).

10. Conclusion

This paper developed a PLC-based multi-sensor condition-monitoring framework for industrial induction motors using electrical current and vibration signals. The principal engineering proposition is that complementary electrical and mechanical evidence can be converted into transparent condition indicators and evaluated through a PLC-based decision layer.

The revised methodology strengthens the original concept by defining the complete measurement chain, sensor-placement discipline, feature extraction equations, normalization procedure, weighted sensor-fusion score, PLC state logic, HMI requirements and experimental validation procedure. Worked calculations demonstrate how current RMS, phase-current imbalance, vibration RMS, crest factor and FFT resolution can be obtained from sampled signals.

The paper deliberately avoids presenting illustrative plots or hypothetical classification metrics as measured results. The next research stage is therefore experimental implementation on a controlled motor test rig or approved industrial equipment. After collection of healthy and controlled-fault datasets, the proposed thresholds, fusion weights and PLC logic can be calibrated and the system can be evaluated using detection rate, false-alarm rate, precision, detection latency and monitoring availability.

The resulting framework provides a practical intermediate step between conventional preventive maintenance and advanced predictive analytics. Once reliable data have been accumulated, future work can investigate additional variables such as temperature, voltage imbalance, speed and power factor, followed by statistical or machine-learning models where they provide measurable benefit.

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Author Note

Takunda Pambayi is an Electrical/Automation Engineering professional and Master's student in Industrial Automation at the Harare Institute of Technology, Zimbabwe. His technical interests include industrial automation, PLC systems, motor drives, condition monitoring, predictive maintenance and energy management.

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